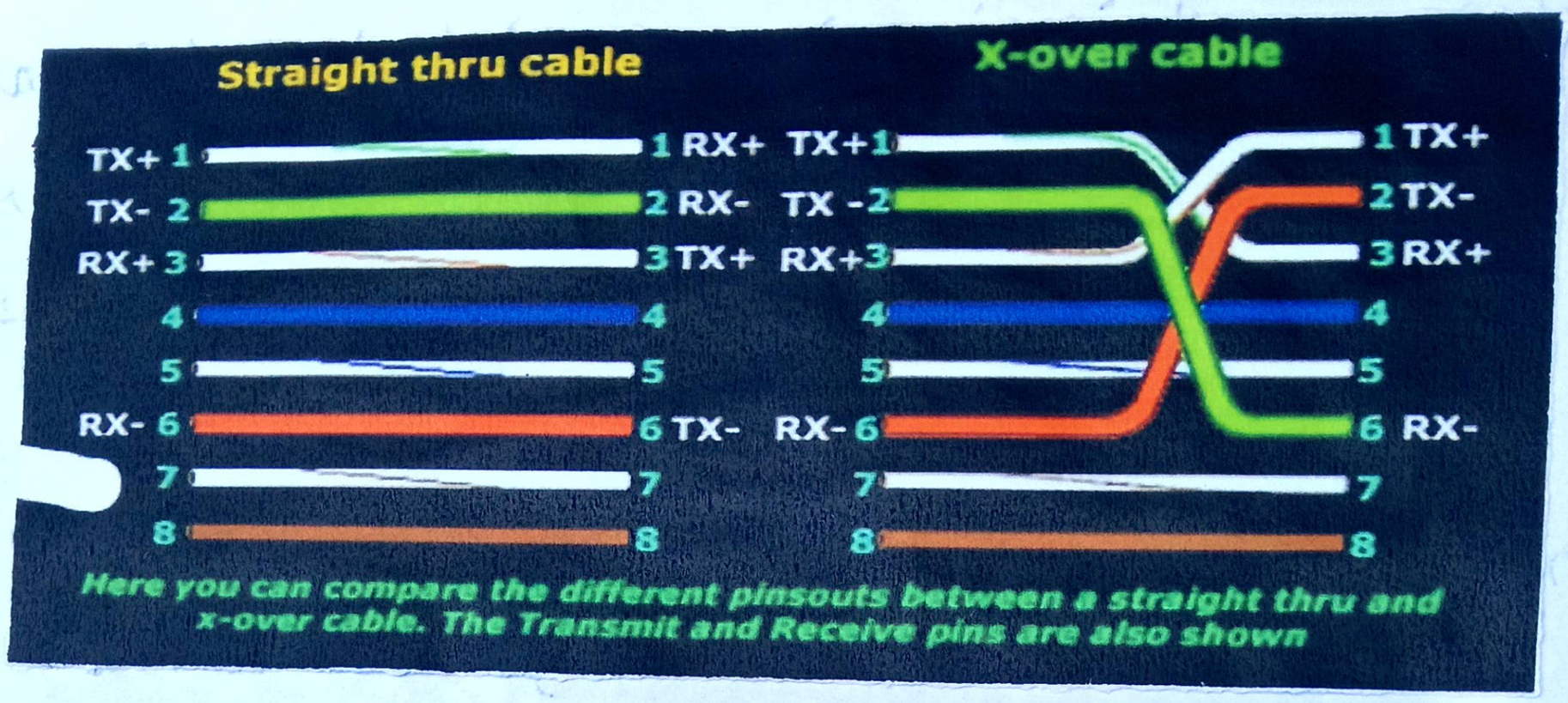

Straight thru cable
X-over cable
TX+ 1
1 RX+
TX- 2
2 RX-
RX+ 3
3 TX+
4
4
5
5
RX- 6
6 TX-
7
7
8
8
TX+1
1 TX+
TX -2
2 TX-
RX+3
3 RX+
4
4
5
5
RX- 6
6 RX-
7
7
8
8
Here you can compare the different pinsouts between a straight thru and x-over cable. The Transmit and Receive pins are also shown


orange solid
white/green stripe
blue solid
white/blue stripe
green solid
white/brown stripe
Clip underneath, pins facing you
Clip underneath, pins facing you
green solid
white/orange stripe
white/brown stripe
brown solid
blue
white

·No: 2

IE:

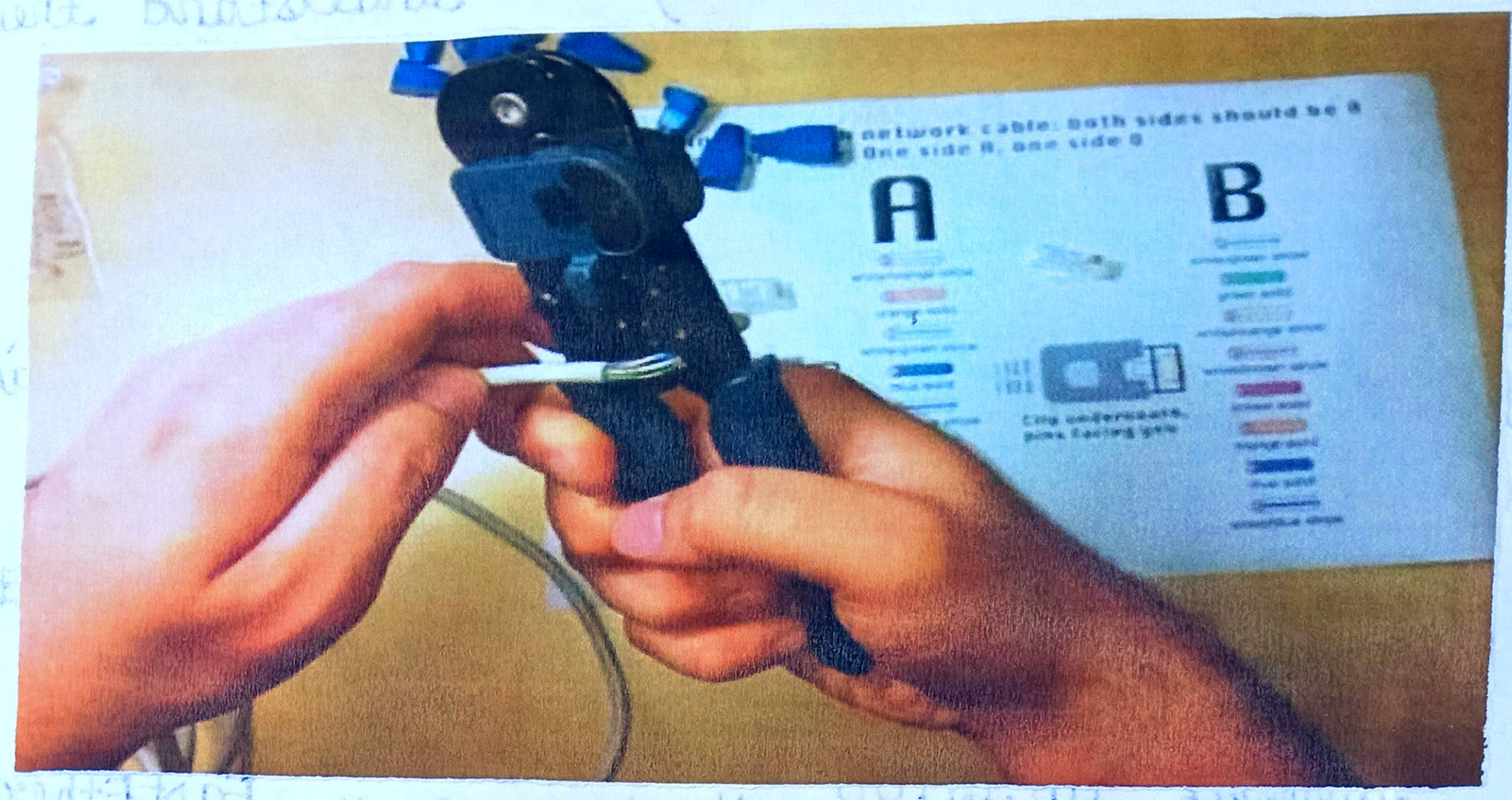
3·7·26

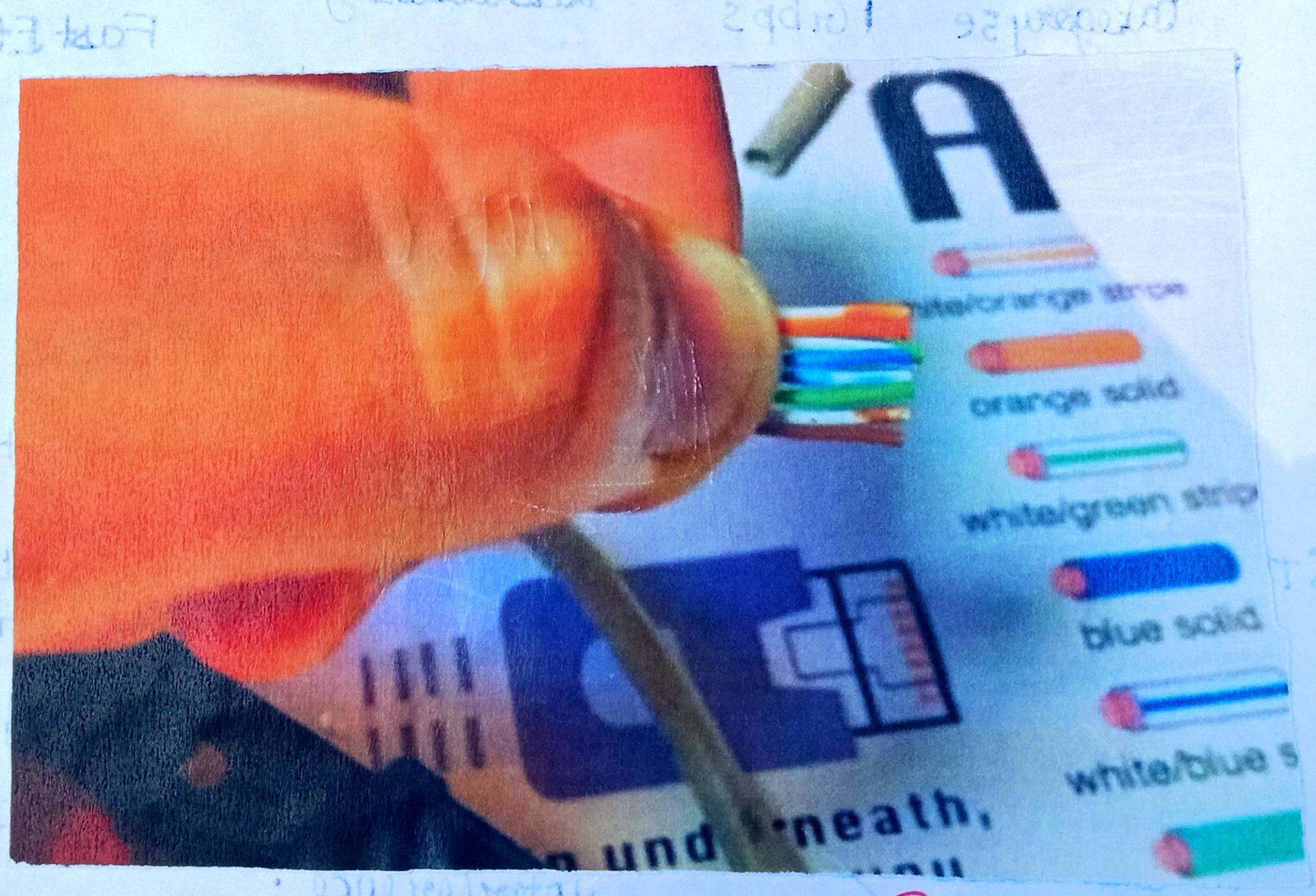
STUDY OF DIFFERENT TYPES OF NETWORK CABLES

AIM:-

To study the different types of network cables used in computer net working and understand their purpose and applications.

Cable type	Category	Maximum Data Transmission	Advantages / Disadvantages	Application / Use
UTP	category 3	10 bps	Advantages • cheapter in cost • Easy to install as they have a smaller overall diameter. Disadvantages • More prone to (EMI) Electromagnetic interferance and noise.	10Base - T Ethernet
	category 5	up to 100 Mbps		Fast Ethernet, Gigabit Ethernet
	Category 5e	1 Gbps		Fast Ethernet, Gigabit Ethernet
STP	category 6, 6a	10 Gbps	Advantages • Shielded. • Faster than UTP. • Less susceptible to noise and interference. Disadvantages • Expensive • Greater installation effort.	Gigabit Ethernet, 10 G Ethernet (5 5m) widely used in data centres.
SSTP	category 7	10 Gbps		Gigabit Ethernet 10G Ethernet (100 m)


network cable: both sides should be A
One side A, one side B
A
B

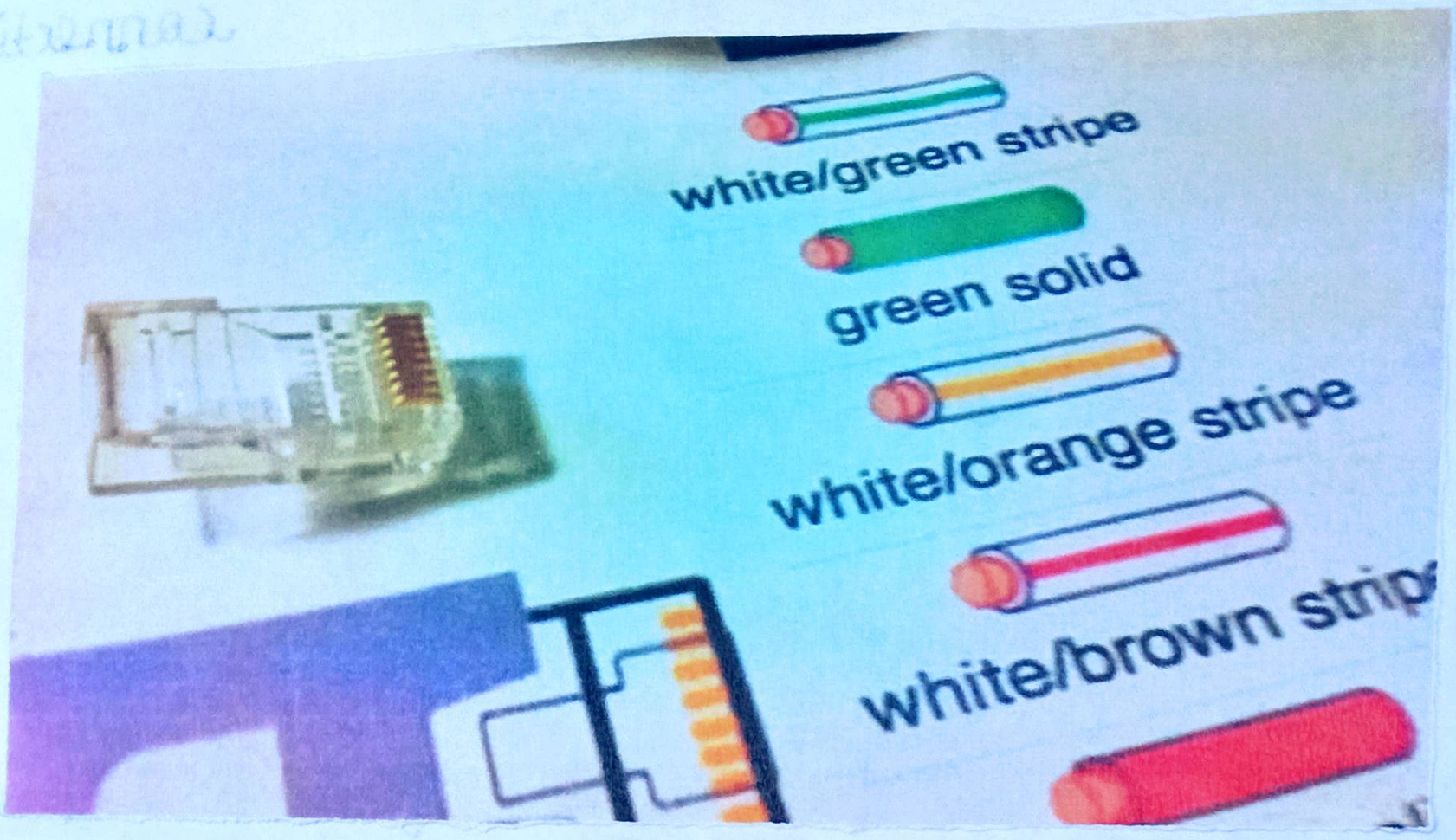

A
orange stripe
orange solid
white/green stripe
blue solid
white/blue s
und

Coaxial cable	RG-6 RG-59 RG-11	10-100Mbps	• High bandwidth • Immune to interference • Low loss bandwidth • Versatile Disadvantages • Limited distance • Cost • Size is bulky.	Speed of signal is 500m Television network High speed Internet connections.
fibre optics cables	Single mode Multi mode	100 Gbps	Advantages • High Speed • High bandwidth • High Security. • Long distance Disadvantages • Expensive • Requires Skilled installers	• Maximum distance of fibre optics cable is around 100 meters.

b) Make your own Ethernet cross-over cable / Straight cable :-

Tools and parts needed :

- Ethernet cabling. CAT5e is certified for gigabit support, but CAT5 cabling works as well, just over short distances.
- Two RJ45 plugs.


white/green stripe
green solid
white/orange stripe
white/brown stripe


A
white/orange stripe
orange solid
white/green stripe
blue solid
white/blue stripe
underneath,
facing you

• optional two plug shields.

Step 1:- To start construction of the device, begin by threading shields onto the cable.

Step 2: Next, strip approximately 1.5 cm of cable shielding from both ends. The crimping tool has a round area to complete this task.

Step 3:- After, you will need to untangle the wires; there should be four "twisted pairs." Referencing back to the sheet, arrange them from top to bottom. One end should be in arrangement A and the other in B.

Step 4:- Once the order is correct, bunch them together in a line, and if there are any that stick out farther than others, snip them back to create an even level. The difficult aspect is placing these into the RJ45 plug without messing up the order. To do so, hold the plug with the clip side facing away from you and have the gold pins facing toward you, as shown.

Step 5:- Next, push the cable right in. The notch at the end of the plug needs to be just over the cable shileding, and if it isn't that means that you stripped off too much shielding. Simply snip the cables back a little more.

Step 6 :- After the wires are securely sitting inside the plug, insert it into the crimping tool and push down.

Step 7 :- Lastly, repeat for the other end using diagram B (to make a crossover cables)/using diagram A (to make a straight through cable.

Result :-

Thus, different type of network cables were Studied successfully, and straigh through and cross over ethernet cables were prepared and tested successfully.